Greenhouse Climate Controller

Match plants to the right climate zone

At a glance: Run the greenhouse controls for a plant that cannot complain, only wilt. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: controlled environments, temperature, humidity, light, plant needs, tradeoffs (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- plant profile cards: one set per team (4, plus 1 spare)
- control dial boards: one per team (4, plus 1 spare)
- tour clue sheets: one per team (4, plus 2 spare)
- clipboards: one per team (4)
- dry erase markers: shared, 1 to 2 per team

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE is needed: the tabletop phase is a dry card-and-board game with no chemicals, soil, or water. For the greenhouse tour, confirm the route and off-limits plants and controls with greenhouse staff, and check the roster for plant or pollen allergies.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Run the greenhouse controls for a plant that cannot complain, only wilt." Take a quick prediction by show of hands.

Phase 2 Prediction

`0:05 - 0:12`

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

`0:12 - 0:18`

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

`0:18 - 0:44`

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Match and set

Teams sort plant cards by need, match each plant to a zone, and set temperature, humidity, and light on the dial board using their tour evidence.

Phase 6 Resolve tradeoffs

Where two plants conflict in a zone, or raising temperature would dry the air too much, each team picks the setting its evidence best supports and notes the tradeoff it accepted.

Phase 7 Score, debrief, cleanup

`last 12 min`

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wipe the dry-erase boards and return all cards, boards, and clue sheets to the bin so the next team has a complete set; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A team places a plant with no supporting tour evidence. Fix: Ask which clue from the tour backs that zone choice; if they have none, send them back to their clue sheet to find or admit the gap before scoring.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to controlled environments.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Temple Ambler Greenhouse

Plant profile cards

Each card says what one plant needs. Match it to the greenhouse zone whose tour readings fit, then defend the placement with a clue from the tour.

Fern

TEMP	cool to warm, 16 to 24 °C (60 to 75 °F)
HUMIDITY	high, 50% or more; likes damp air
LIGHT	shade or bright indirect; never direct sun

WANTS Damp air and shade, like a forest floor.
AVOID Direct sun and dry air crisp the fronds.

Moth orchid

TEMP	warm, 24 to 29 °C (75 to 85 °F) by day, above 16 °C (60 °F) at night
HUMIDITY	humid, 50 to 80%, with some air movement
LIGHT	medium; bright but filtered, no direct beam

WANTS Warm, humid air and bright, filtered light.
AVOID Full direct sun scorches the leaves (the orchid-loves-sun myth).

Cactus

TEMP	warm days, big day-to-night swing, 24 to 34 °C (75 to 90 °F)
HUMIDITY	low; dry air and fast-draining soil
LIGHT	very high; full, direct sun

WANTS Hot, bright, dry air like a desert shelf.
AVOID Damp, shady air rots the roots.

How to read a card

The thick black bar on each scale is the plant's happy range.

TEMP
10 to 40 °C.

HUMIDITY
0 to 100%.

LIGHT
shade on the left to full sun on the right. A plant in the wrong zone tells you how it suffers: too dry wilts, too bright scorches, too damp molds.

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Control dial board

Set each dial for the plant you are placing, then record where each plant goes and the tour clue that backs it.

TEMPERATURE (°C)

10 20 30 40

mark your setting with a dry-erase pen

HUMIDITY (% RH)

0 25 50 75 100

mark your setting with a dry-erase pen

LIGHT

shade filtered bright full sun

mark your setting with a dry-erase pen

Placements: defend each one with a tour clue

PLANT	ZONE YOU CHOSE	TEMP	HUMIDITY	LIGHT	TOUR CLUE THAT SUPPORTS IT
Fern					
Moth orchid					
Cactus					

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Tour clue sheet

As you tour, record what you see, feel, and hear for each zone: warm or cool, damp or dry, bright or shaded, plus any reading the docent points out. Back at the table this is your evidence for matching plants to zones.

GREENHOUSE ZONE	TEMPERATURE	HUMIDITY (DAMP/DRY)	LIGHT (SHADE/BRIGHT)	WHAT THE DOCENT SAID / EVIDENCE
Main greenhouse				
Hoop house				
Shaded bench				
Bright bench				

Tip: you do not need exact numbers. "Warmer and more humid than the hoop house" is good evidence. Use the control-system or weather-station readout if the docent shows one.